

EVENTWEB: A Unified Campus Event Discovery, GIS Mapping, and Ticketing Infrastructure

Architectural Design, Distributed Cloud Backend, Spatial GIS Integration, and Transactional Notification Pipeline

Target Institution: KNUST, Kumasi, Ghana	Domain: Web GIS & Cloud Software Eng.	Backend & Auth: Supabase (PostgreSQL + RLS)	Email Dispatch: Resend (Edge Functions)
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1. Executive Summary & Problem Statement

Campus life at Kwame Nkrumah University of Science and Technology (KNUST) features hundreds of student-driven and faculty-organized activities per academic semester—spanning academic symposiums, society dinners, sports derbies, religious retreats, and entrepreneurship bootcamps. Historically, publicity has been scattered across ephemeral WhatsApp status updates, unstructured broadcast groups, and physical paper posters pasted over campus notice boards. This fragmentation leads to information decay, low turnouts, double-booked halls, and no centralized check-in records.

EVENTWEB unifies campus events into a single digital platform: a live-updating discovery board categorized by five distinct activity tracks (Academic, Social, Sports, Religious, Business), an interactive campus GIS cartography map, an automated pass generation engine with door verification codes (EW-XXXX-XX), and a verified society organizer console.

2. System Architecture & Component Model

EVENTWEB employs a modern decoupled architecture balancing immediate client rendering with resilient cloud persistence and serverless communication.

```
+-----+ | EVENTWEB PLATFORM |
+-----+ | STUDENT CLIENT | ORGANIZER
CONSOLE | | * 5s Auto-Rotating Poster Board | * Verified Society Authentication | | * Category Filtering &
Search | * Flyer Upload (Supabase Storage) | | * Interactive KNUST GIS Map | * Capacity & Tier Configuration |
| * Instant Registration & Code Gen | * Real-Time Attendee Tracking |
+-----+ | | v v
+-----+ | SUPABASE CLOUD INFRASTRUCTURE |
| +-----+ +-----+ +-----+ | | PostgreSQL Database | | Supabase
Auth | | Storage Buckets | | | (RLS + Triggers) | | (JWT & Sessions) | | (event-posters) | | |
+-----+ +-----+ +-----+ | | | v Database Webhook | |
+-----+ | | Supabase Edge Function (Deno
Runtime) + Resend Email API | | | * Automated Ticket Email with Reference Code & Calendar Details | | |
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```

3. Technology Stack & Technical Justifications

Subsystem	Technology	Engineering Rationale
Frontend Core	HTML5 / Vanilla JS	Eliminates virtual-DOM overhead, delivering instantaneous First Contentful Paint.
Design System	Modernist CSS (1px rules)	Zero border-radius, clean 1px hairlines, and strict WCAG AA contrast compliance.
GIS Engine	Leaflet.js + OSM Tiles	Client-side campus cartography with custom coordinate bounding boxes.
Database	Supabase (PostgreSQL 15)	ACID transactions, custom ENUMs, JSONB tiers, and native Row-Level Security.
Auth & Storage	Supabase Auth & Storage	JWT organizer sessions and CDN bucket storage for society flyers.
Email Delivery	Resend + Edge Functions	Serverless execution for instant pass delivery and automated door reminders.

4. Database Schema & Data Integrity

The relational schema is normalized into three core entities: organizers, events, and registrations. Row-Level Security (RLS) protects unauthorized modification, while a PostgreSQL trigger eliminates overbooking race conditions.

```
[ ORGANIZERS ] 1:N [ EVENTS ] - id (UUID, PK, Auth) -----> - id (UUID, PK) - org_name (TEXT) - organizer_id (UUID, FK) - org_type (TEXT) - title (TEXT) - email (TEXT) - category (ENUM) - is_verified (BOOL) - venue (TEXT) - date (DATE), time (TEXT) - capacity, reg_count (INT) - poster_url (TEXT) - options (JSONB) - status (ENUM) | 1:N v [ REGISTRATIONS ] - id (UUID, PK) - event_id (UUID, FK) - code (TEXT, Unique) - attendee_name, email (TEXT) - remind_opt_in (BOOL)
```

4.1 Concurrency & Registration Counter Trigger

To ensure live registration counts remain atomic and immune to client tampering, a database trigger automatically increments `reg_count` on the target event immediately upon successful registration insertion:

```
CREATE OR REPLACE FUNCTION public.handle_new_registration() RETURNS trigger AS $$ BEGIN UPDATE public.events SET reg_count = reg_count + 1 WHERE id = new.event_id; RETURN new; END; $$ LANGUAGE plpgsql SECURITY DEFINER;
CREATE TRIGGER on_registration_created AFTER INSERT ON public.registrations FOR EACH ROW EXECUTE PROCEDURE public.handle_new_registration();
```

5. Interactive GIS Campus Cartography

The spatial mapping module embeds KNUST campus coordinates into a custom Leaflet.js canvas, resolving venue locations accurately:

- **Great Hall:** 6.67475 N, -1.57220 W (Central Assembly & Keynote Hall)
- **Paa Joe Stadium:** 6.67780 N, -1.56950 W (Athletics, Tournaments, Outdoor Fairs)
- **College of Science Auditorium:** 6.67350 N, -1.56650 W (Symposiums & Lectures)
- **Prempeh II Library:** 6.67510 N, -1.57180 W (Academic Workshops & Book Clubs)
- **KNUST Interdenominational Church:** 6.68508 N, -1.57270 W (Services & Fellowships)
- **Brunei Sports Complex / Campus:** 6.67644 N, -1.57343 W (Night Markets & Festivals)

Architectural Drafting Callouts: Markers feature a 2px coordinate stem with an interactive flag displaying venue numbers and category color bars. Clicking a pin highlights events in the list; selecting a list venue pans the map smoothly.

6. Front-End Interaction & Modernist Design Language

- **5-Second Hero Autoplay:** The 3D poster stack rotates automatically every 5000ms. If the student clicks, drags, or browses manually, the timer is cleared and restarts a fresh 5-second countdown from that moment.
- **1px Seamless Hairlines:** Replaced brutalist borders with refined 1px strokes (--ew-rule: 1px). In dark mode, borders switch to #29324a to blend seamlessly against dark grounds.
- **Modal Isolation:** When an event registration sheet is opened, background poster rotation is paused to preserve user concentration.

7. Automated Transactional Pass Pipeline (Resend)

Upon registration insertion, Supabase triggers an Edge Function (Deno) that interfaces with Resend: attendees receive a structured email pass containing their unique code (e.g. EW-4471-KQ), venue directions, and Google Calendar integration. API credentials remain encrypted in Supabase environment secrets.

8. Conclusion & Academic Defense Summary

EVENTWEB bridges modern web engineering with campus community needs, replacing chaotic publicity channels with an institutional-grade, aesthetically stunning platform. Its decoupled architecture guarantees sub-second responsiveness, data privacy via RLS, and a scalable foundation for tertiary event discovery across West Africa.